20 $d = \frac{1}{8}c^3$

$c = 10.9$ correct to 3 significant figures.

By considering bounds, work out the value of d to a suitable degree of accuracy.
Give a reason for your answer.

$10.85 \leq c < 10.95$ from 3 significant figures.

Lower $d = 10.85^3/8 \approx 159.6611$.

Upper $d = 10.95^3/8 \approx 164.1165$.

Both bounds round to 160 to nearest ten.

So $d = 160$ to a suitable degree of accuracy.

4 marks total
Bounds for c
Cube both / divide by 8
d bounds 159.66–164.12
Both give 160 to nearest 10

(Total for Question 20 is 4 marks)